

Project Topics

Machine Learning

1. Imbalanced tabular data poses a persistent challenge in machine learning, as classifiers often underperform on the minority-class data when trained on skewed data. Synthetic minority data generation is a standard approach, with traditional interpolation-based methods offering efficiency but limited representational capacity and being inapplicable under severe imbalance due to their reliance on existing data points. Recent approaches that employ large language models (LLMs) address this limitation by generating synthetic samples informed by contextual knowledge. The project may extend the idea to non-convex data distributions and develop new method for generating synthetic samples.

Papers:

Imbalanced Tabular Data Synthesis via LLM-Seeds and Interpolation

2. Graph Neural Networks (GNNs) have achieved remarkable success across various graph analysis tasks. However, in real-world scenarios, graphs typically exhibit a class-imbalanced distribution, where a few classes account for the majority of nodes. This distribution leads to a bias toward majority classes and degrades the model's discriminative ability on minority classes. It is a promising solution to leverage the numerous unlabeled samples to rebalance the class distribution. The project aims to improve upon existing methods to address class imbalance in node classification in graphs.
 - a. The project may explore possibility of adversarial generation of nodes
 - b. May explore the possibility of graph clustering in GraphUAT framework
 - c. May explore applications in imbalanced node classification in specific domains like medical science, traffic analysis, EEG based predictive tasks

Paper:

- i. GraphUAT: An uncertainty-driven pseudo-labeling method for class-imbalanced node classification
3. This project may address class imbalance in machine learning classification tasks, where data distribution is unequal across classes. Existing resampling methods, which balance classes through random oversampling or under sampling, can introduce noise or information loss. The project may explore the 'Uncertainty guided oversampling' method and explore the possibility to improve fundamental challenges of imbalanced classification like:
 - a. How to generate minority samples close to decision boundary without increasing class overlapping?
 - b. How to handle intra-class imbalance?
 - c. Can we leverage unlabelled data to generate synthetic minority samples?

Paper:

- i. Uncertainty guided oversampling: A classifier-specific approach for effective handling of class imbalance in machine learning

4. Depression is a significant global health issue with increasing prevalence. Current diagnostic methods rely on subjective observations and questionnaires, often resulting in underestimation of the condition and insufficient treatment. Speech recordings and their automatic transcriptions were processed using state-of-the-art pretrained transformer-based models to generate embeddings for further classification. However, the latent space is highly overlapped. The project may explore the possibility of;
 - a. Addressing class overlapping by using contrastive learning
 - b. Addressing class imbalance by latent space oversampling

Paper:

- i. Detecting depression through speech and text from casual talks with fully automated virtual humans
5. Multimodal learning aims to integrate diverse data sources to capture more comprehensive information about things, thus enhancing perception and understanding of the real world. However, inherent discrepancies between different modalities often lead to imbalanced optimization during multimodal learning, hindering performance improvement. The project may extend the idea further for multimodal models that involves medical data.
 - a. It may develop methods to address class imbalance in such multimodal setup

Paper:

- i. Deep Information-Balanced Multimodal Learning
6. Multimodal models trained on centralized medical datasets frequently fail when deployed in local clinics due to differences in equipment manufacturers, specifications, or patient demographics (out-of-distribution shifts). The project may Investigate Contextual Control Anchor Mechanisms for single-domain generalization. This allows a medical vision model to adapt to a completely unseen hospital dataset at test-time using only a few unlabeled images.

Paper:

- i. Experience with Single Domain Generalization in Real World Medical Imaging Deployments
7. Traditional multimodal FL assumes every hospital collects the exact same data types (e.g., all have EHR + MRI). In reality, Hospital A might only have text reports, while Hospital B only has imaging scans. Standard architectures fail when clients have completely missing modalities. Create a Federated Information-Balanced Fusion Network for Missing Modalities. May adapt the core fusion mechanics from the paper “*Deep Multimodal Information-Balanced Learning (IBML)*” into a federated server aggregation pipeline to handle asymmetrical sensor inputs across distributed nodes.
 8. Traditional FL enforces a strict structural constraint: every client must run the exact same neural network architecture (e.g., all ResNet-50 or all ViT-B). In diverse IoT or cross-organizational networks, Client A might only support an ultra-light MobileNet,

while Client B runs a massive Transformer. Standard parameter aggregation (FedAvg) collapses here because weights cannot be matched matrix-for-matrix. Investigate Agnostic Logit-Space Federated Co-Distillation. Instead of aggregating model weights, clients transfer highly compressed, non-invertible output token probabilities (logits) onto a shared, unlabelled public proxy dataset hosted at the server, allowing completely heterogeneous client architectures to learn from one another seamlessly.

Paper:

- i. GeFL: Model-Agnostic Federated Learning With Generative Models
9. Cross-entropy loss inherently collapses minority class features into tight, fragile spaces right next to dominant majority class boundaries. While vanilla Supervised Contrastive Learning (SupCon) improves feature clustering, it pushes all classes equally, allowing large classes to visually swallow rare classes that possess high intra-class variance. A Density-Aware Angular Margin Contrastive Loss can be developed. This loss function adaptively widens the angular distance (margin) in the embedding space based on both class scarcity and local cluster density. Rare classes are granted a wider safety buffer on the hypersphere, preventing majority samples from encroaching on their feature territory.

Papers:

- i. Adapting Supervised Contrastive Learning to Binary Imbalanced Datasets
 - ii. Equilibrium Contrastive Learning for Imbalanced Image Classification
10. Long-tailed class distributions are pervasive in multi-class medical datasets and pose significant challenges for deep learning models which typically underperform on tail classes with limited samples. This limitation is particularly problematic in medical applications, where rare classes often correspond to severe or high-risk diseases and therefore require high diagnostic accuracy. In this context explore the paper “Synthetic Data Generation for Long-Tail Medical Image Classification: A Case Study in Skin Lesions” to improve its performance.
11. Recent advances in Multimodal Large Language Models (MLLMs) have shown promising results in integrating diverse modalities such as texts and images. MLLMs are heavily influenced by modality bias, often relying on language while under-utilizing other modalities like visual inputs. The project may explore ideas to mitigate modality bias in MLLMs.

Paper:

- i. MLLMs are Deeply Affected by Modality Bias

Dataset (Medical Data):

1. <https://github.com/beamandrew/medical-data>
2. <https://archive.physionet.org/physiobank/database/mimiccxr/>
3. <https://stanfordmlgroup.github.io/competitions/chexpert/>
4. <https://openi.nlm.nih.gov/faq>